

Evaluation Framework for Multimodal Language Models Applied to Geospatial Open Data in Agriculture

Modular architecture

Satellite



Vegetation indices

Orthophoto



High-res imagery

RAG



Knowledge base

Dataset construction

- (1) Parcel selection
- (2) LU-related query elaboration
- (1) Response generation

143 samples

Methodology stages

Broad

Ablation

Panel of judges



Hybrid Human-in-the-Loop

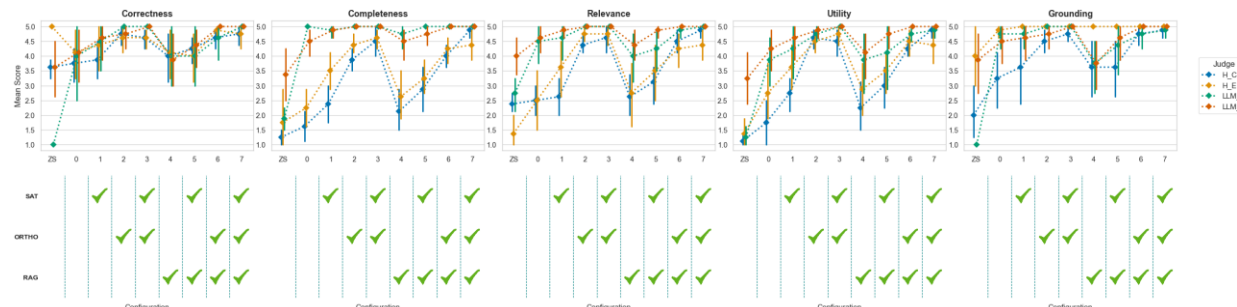
Metrics

- ✓ Correctness
 - ✓ Completeness
 - ✓ Relevance
 - ✓ Utility
 - ✓ Grounding
- 1-5 Likert

Analysis & key findings

Collaborative game theory framework for marginal contributions

SHAPLEY Values



- Validated architecture
- Visual context crucial
- Marginal contributions successfully quantified
- Reproducibility
- Dataset is available